

SHARUGESAN R

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PROFESSIONAL SUMMARY

Recent B.Tech AI & DS graduate from PSNACET with a CGPA of 8.06, specializing in full-stack AI application development. Built AramAI, a multilingual legal assistant using React, FastAPI, LoRA fine-tuning, and Groq API, serving marginalized communities in Tamil Nadu. Experienced in AIoT systems, cloud platforms, and end-to-end software development. National Finalist at K-Impact 2025.

SKILLS

Programming Languages & Tools — Python, Java, C, JavaScript, HTML, CSS, SQL

Frameworks & Libraries — React.js, FastAPI, NumPy, Pandas, Scikit-learn, Matplotlib

AI/ML — LoRA Fine-Tuning, HuggingFace (Transformers, PEFT, TRL), LLM Integration (Groq/LLaMA), RAG, ChromaDB, Sentence Transformers

Tools & Platforms — Git, GitHub, Google Colab, AWS, GCP, Arduino, Raspberry Pi

Databases — MySQL and SQL

Core Concepts — DSA, OOP, REST APIs, Problem Solving, Debugging

PROJECTS

AramAI – Multilingual AI Legal Assistant for Marginalized Communities  01/2026 – 05/2026

- Built a full-stack AI web application using React.js (frontend) and FastAPI (backend) to provide free multilingual legal guidance in English and Tamil to marginalized communities.
- Fine-tuned TinyLlama 1.1B using LoRA on 21,948 Indian Supreme Court case pairs; trained only 0.36% of parameters (2.25M/617.86M) achieving training loss of 0.657.
- Integrated Groq API (LLaMA 3.3 70B) for 2-second responses; implemented Tamil voice input/output using Sarvam AI (saarika:v2.5 STT + bulbul:v2 TTS).
- Features include conversation memory, complaint letter drafting, PDF generation (ReportLab), and light/dark theme with fully responsive mobile design.
- Model hosted on HuggingFace: huggingface.co/sharugesanr/aramAI-legal 

An AIoT-based Vehicle Gas Emission Detection and Reporting System  08/2024 – 05/2026

- Built a smart system using MQ sensors and Arduino to monitor vehicle emissions in real time, displaying data on an LCD and updating a web dashboard. Alerts are sent to users and authorities when emission levels exceed limits.
- National Finalist at K-Impact 2025, evaluated by KPIT Technologies - scored 51/60 on feasibility, innovation, and implementation potential.
- Patent application filed (application number awaited).

Web Application Designer (SIH Project)

09/2024

- Designed a web app for CO₂ emission analysis from coal mines under Smart India Hackathon (SIH), offering insights for carbon management through emission estimates, carbon sink mapping, and reduction strategies.
- Built using HTML, CSS, and JavaScript as part of Smart India Hackathon 2024.

INTERNSHIP

Full Stack Development Intern

06/2025 – 07/2025

Eagle-HiTech Softclou Private Limited

Chennai

- Developed patient registration and appointment scheduling modules for a Hospital Management System using HTML, CSS, JavaScript, and MySQL.
- Collaborated with a team to improve user experience and ensure smooth system performance.

EDUCATION

B.Tech AI&DS

2022 – 2026

PSNA College of Engineering and Technology

Dindigul

- Current CGPA:8.06

HSC

2021 – 2022

Syed Ammal Boys Metric Higher Secondary School

Ramanathapuram

- Percentage:89.16%

SSLC

2019 – 2020

Syed Ammal Boys Metric Higher Secondary School

Ramanathapuram

- Percentage:79%

ACHIEVEMENTS

K-Impact 2025 – National Finalist

08/2025

- Selected as National Finalist for the AIoT-based Vehicle Gas Emission Detection project. Evaluated by KPIT Technologies — scored 51/60.

Coding Contest - 2nd Runner-up

04/2024

- Secured 2nd Runner-up in the Coding Contest during our department's symposium.

PSNACET

PATENT

An AIoT-based Vehicle Gas Emission Detection and Reporting System

2025 – Present

- Patent application filed — Application number awaited.
- Co-inventors: Sharugesan R., Edwin Joel P., Nithya R., Abirami K.

CERTIFICATIONS

R Programming — Spoken Tutorial Project, IIT Bombay (Nov 2025)

PHP and MySQL — Spoken Tutorial Project, IIT Bombay (Feb 2025)

Cloud Security Fundamentals — Palo Alto Networks (Feb 2025)

Linux CLI Basics — Cybrary (Feb 2025)

Business Analysis Basics – Simplilearn (Jan 2025)

AWS Badge — AWS Cloud Club (Nov 2024)